BLOG POST

HOW RESTORING BOREHOLES IS TRANSFORMING HEALTH, COMMUNITIES AND CLIMATE RESILIENCE IN UGANDA

For many people, turning on a tap is an everyday routine. For thousands of families in rural Uganda, however, getting safe water meant beginning the day with a long walk. In the districts of Lyantonde, Kabarole and Kalungu, women and children carried empty jerry cans for kilometres to collect water, often from unsafe sources. Once home, many families had to boil water over firewood, spending even more time gathering fuel. The cost was never just water. It was time lost, missed school days and the greater risk of waterborne diseases, indoor air pollution from the firewood and a changing ecosystem due to deforestation. A quieter cost accumulated in the lives of women and girls, whose time and safety were spent covering a gap that infrastructure was meant to fill.

A teenager collecting firewood in Kalungu District, Uganda

The Problem

On paper, these districts had boreholes; but in practice, non-functional with broken pumps, no one trained to fix them and no funds set aside for upkeep. So, households did what they had to do when the system fails them; they found the nearest source, safe or not and they made do.

The Change

Recognising these challenges, the Sustainable Climate Impact Fund (SCIF) launched an initiative to restore access to safe and reliable water. Instead of building entirely new infrastructure, the project focused on repairing what communities had already lost. Over 70 non-functional boreholes were rehabilitated across villages, schools, and healthcare facilities, while three solar-powered motorised water systems were installed to serve areas where demand was highest. For many families, this meant that safe water was finally available much closer to home.

Motorised Waterpoint in Mukoko, Kalungu District

SCIF understood that restoring boreholes was only the first step. Across many parts of Africa, water systems often fail not because they cannot be repaired, but because communities lack the skills, resources, or support to keep them working. When this happens, families are forced back to unsafe water sources, putting their health and well-being at risk.

To prevent this, the project placed communities at the centre of the solution. Water User Committees were trained and equipped to manage the day-to-day operation of the boreholes, while district hand pump mechanics trained to carry out repairs when needed. Committee members also learned basic financial management and accountability, helping them manage maintenance costs and respond quickly to problems

Mechanics Repairing Pump

The project also recognised that improving access to safe water alone would not be enough to improve health. Working with WaterAid Uganda, SCIF delivered hygiene awareness campaigns through community meetings and household visits. Families learned about safe water storage, sanitation and hygiene practices that reduce the spread of disease. The district government was also brought in early, so that ownership didn't end when the project did.

Changing behaviour, however, took longer than repairing infrastructure. While a broken borehole could be fixed within weeks, building lasting habits required continuous engagement and trust. The experience highlighted an important lesson: lasting improvements in public health depend not only on infrastructure but also on empowering people with the knowledge to use it safely.

The changes were soon felt across the community. Today, more than 200,000 people benefit from the rehabilitated boreholes, the solar-powered systems provide safe water to more than 6,000 people every day and 276 committee members and mechanics have acquired maintenance skills.

Phiona Nangoozi collecting water from a SCIF rehabilitated borehole in Katungulu

Health centres that once asked labouring women to carry their own 20-litre jerrycan of water no longer have to, reducing physical strain and improving dignity during childbirth. The shift away from boiling water over firewood has also helped avoid an estimated 79,000 tonnes of CO_2e, verified under the Gold Standard framework; a reminder that safe water and climate action were never separate problems.

Night Proscovia, a Water User Committee Chairperson, put it simply:

"The borehole has been a blessing. Especially for us as women and for the children too, as they have been travelling long distances to fetch water and this affected their studies."

Community Water Access Point

What didn't go as plan

As much as hygiene campaigns reached households, shifting everyday habits on water handling and storage took far more repeated engagement that the project timeline allowed for. And even as boreholes were restored, the land around them was left exposed. No dedicated funding existed to protect the catchment areas feeding these water sources, leaving the very groundwater the project depended on vulnerable to the next round of degradation.

What policymakers and programmes should take away

Lasting impact depends not only on restoring infrastructure but also on empowering communities to manage and maintain it. Investing in local skills and technical capacity keeps water systems functioning, strengthens resilience and reduces dependence on external support. At the same time, protecting catchments and groundwater recharge areas is as important as fixing the tap, and that means folding catchment protection into district climate action plans from the very beginning, not as an afterthought once the pipes are already laid.

This story from rural Uganda shows that some of the most effective climate and health solutions begin with meeting people's basic needs. A functioning borehole can improve health, keep children in school, reduce the burden on women, protect forests and strengthen community resilience. When people are given the knowledge, skills and ownership to sustain these gains, the benefits extend far beyond access to safe water, creating healthier and more resilient communities for generations to come.

Case study: Sustainable Climate Impact Fund.
Shared as part of the CAPCHA case study collection by John Starmer, SCIF Safe Water Uganda.

Compiled by the CAPCHA Team: Ann Irungu, Virginia Wamboi and Maria Nailantei